Analytical Documentation

Student Planner

Desktop Productivity Application
Software Engineering
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1. Business Description (Business Requirements)

1.1 Context and Product Origin

The modern academic environment is characterized by a high degree of information fragmentation, which poses a significant organizational burden for students. Everyday challenges of study planning are hindered by the necessity to use multiple unintegrated channels: official university systems, e-learning platforms, mailboxes, messengers, and traditional notes.

As a result, key data becomes fragmented, and managing it becomes inefficient. Deadlines for exams, quizzes, and project submissions are recorded in external calendars; complex grading rules and grade weights get lost in multi-page PDF syllabi. On the other hand, tracking ongoing progress, such as collecting activity points ("pluses") or monitoring the critical limit of allowed absences, is often handled informally in notebooks or relies entirely on the student's unreliable memory. Such organizational chaos leads to unnecessary stress, missed deadlines, and difficulties in strategic semester planning.

The **Student Planner** project is designed to address these challenges. Its main objective is to create a centralized, intuitive "command center" for every student in the form of a lightweight and fast desktop application. The application aims to integrate all mentioned areas into one place, taking over the burden of remembering limits and deadlines, which will allow the user to focus exclusively on acquiring knowledge.

A key architectural assumption guiding the origin of the project was to make the user independent of internet connectivity issues and server outages at the university (which often experience overloads during busy exam periods). Thanks to the use of a built-in local SQLite database, the application works fully offline. This solution guarantees continuous and immediate access to the most important data without delays, while ensuring full privacy - all information about grades and progress is stored exclusively on the user's physical drive.

1.2. Business Objectives

From the client's perspective (ultimately the student), the software aims to achieve the following business objectives:

1. **Increasing on-time course completion (time management).** The application is designed to prevent forgetting key events (quizzes, exams) through a built-in notification center in the interface and a clear countdown of the time remaining until the exam session.
2. P**rotection against academic dismissal (monitoring rigor).** Providing a tool that allows for precise absence tracking (taking into account upper limits from the syllabus), giving the student a safety buffer against exceeding the allowed number of absences.

3. **Progress visualization and grading transparency.** The system should allow recording every partial activity (points, pluses) against maximum values (e.g., max points for quizzes defined per course). This provides the student with precise information about their current passing percentage (progress).

4. **Knowledge consolidation.** Storing class notes assigned directly to a specific day in the calendar and general course notes within a single environment.

5. **Internationalization.** The ability to instantly switch the interface language (Polish / English), which expands the potential user base to students in international exchange programs (e.g., Erasmus+).

2. Requirements Translation (Requirements Traceability)

The table below demonstrates how the identified Business Requirements (BR) were translated into User Requirements (UR) and ultimately into implemented Functional Requirements (FR) for all defined system modules.

Module	Business Requirement (BR)	User Requirement (UR)	Functional Requirement (FR)
1. Absence Counter	Protection against exceeding the allowed limit.	The user wants to manually check off each class absence and know their limit.	- The "Course Details" module provides a mechanism for manual modification of the absence count through dedicated increment and decrement state functions. - The "Rules Repository" module processes saved values and generates a visual element (a progress bar) that visualizes the degree of utilization of the allowed limit in real time.

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2. Adding Courses	Consolidation of the semester structure.	The user wants to configure course details, its duration, and grading rules.	- The system provides a multi-field course registration form allowing the definition of key data, including: course name, instructor details, grading rules, allowed absence limit, maximum number of points/pluses, and course timeframes (start and end dates). - The module allows configuring recurring classes. The system allows assigning repeatable class blocks to selected days of the week (using a multi-select mechanism), combined with a precise specification of the start time and total duration.

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3. Schedule & Exams View	Time and deadline management.	The user wants to browse their schedule for a specific week with marked events.	- The main system view (dashboard) presents an interactive 7-day calendar equipped with a navigation (scrolling) mechanism between successive/previous weeks. - Recurring classes are visualized as blocks (tiles) whose height is dynamically scaled and directly proportional to the duration of a given event. - High-priority events (exams, tests) are visually highlighted using a dedicated accent color, and all items within a given day are automatically sorted chronologically.
4. Session Countdown Widget	Maintaining time discipline.	The user wants to know everywhere in the app how many days are left until the session starts.	- An interactive module implemented in the sidebar allows the user to define the target start date of the exam session. - The system calculates the remaining time to the set deadline in real time and presents the result as a dynamic countdown accurate to one second.

Module	Business Requirement (BR)	User Requirement (UR)	Functional Requirement (FR)
5. Class Notes	Saving subject-related knowledge every day.	The user wants to click on classes on a given day and make a quick note for them.	- Interaction with an event tile in the calendar view triggers a modal window enabling preview and editing of the assigned daily note. - The system enforces a data integrity rule: exactly one note can exist for a given course within a single calendar day. - Saving content results in an immediate update of the calendar interface, where the tile is marked with a dedicated visual indicator signaling the presence of a note.
6. Rules Repository	Transparency of grading rules.	The user wants to have instant insight into the grading rules of all courses and the absence status.	- The "Rules Repository" view presents a summary of all passing conditions defined during initial course registration in the form of a clear tile grid. - Each tile dynamically displays the "Grading Rules" field (filled out during initial course registration), as well as the current course status and a visual absence limit utilization indicator.

Module	Business Requirement (BR)	User Requirement (UR)	Functional Requirement (FR)
7. Progress Tracker	Progress visualization.	The user wants to add earned points or pluses from classes and see what percentage is missing to the maximum.	- In the "Progress Tracker" tab, the system provides interactive counters (operated via increment and decrement buttons) for recording current activity. - The system sums up the entered values in real time, verifying them against the defined upper limit for a given course. - Data is visualized using a dynamic progress bar.
8. Notifications	Minimization of missed deadlines.	The user wants to have quick insight into upcoming event deadlines (exams, quizzes, projects) to minimize the risk of missing them, without leaving the app or receiving distracting system notifications.	- On the top app bar (regardless of the displayed view), the system constantly presents information about the nearest upcoming event. - Upon clicking the bell icon, the system generates and displays a view containing a list of scheduled events. - Events are automatically sorted by date and time.

Module	Business Requirement (BR)	User Requirement (UR)	Functional Requirement (FR)
9. Course Status	Immediate course identification.	The user wants to manually flag a course as passed, in progress, at risk, or failed.	- The system allows assigning and editing academic status (e.g., in progress, passed) for each course from its details view. - The defined status is dynamically mapped to a color in the user interface. - This change is immediately reflected on tiles representing individual courses in the "My Subjects" grid view.
10. General Note	General data repository.	The user wants to save permanent info like Teams meeting links or lecturer email.	- The course details view contains a dedicated text field for notes. - Entering content and approving it with the "Save changes" button results in the immediate creation (or update) of the corresponding record in the relational SQLite database.

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11. Language Switching	Platform internationalization.	The user wants to operate the application in English or Polish without restarting.	- The button in the Top Bar (PL/ENG) toggles the application state. - The application supports dynamic multilingualism based on an external resource file (JSON). - The system utilizes a property binding mechanism, which allows for an immediate language change of the entire interface without needing to restart the application.